Proving lines parallel

1

a The diagram contains parallel lines. The marked angles are supplementary and consecutive interior angles so the lines are parallel by the converse of the consecutive interior angles theorem.

b The diagram does not contain parallel lines. The marked angles are consecutive interior angles, but not supplementary.

c The diagram contains parallel lines. The marked angles are conguent and corresponding so the lines are parallel by the converse of the corresponding angles postulate.

d The diagram does not contain parallel lines. The marked angles are corresponging angles, but not congruent.

e The diagram contains parallel lines. The marked angles are congruent and alternate interior angles so the lines are parallel by the converse of the alternate interior angles theorem.

f The diagram does not contain parallel lines. The marked angles are alternate interior, but not equal.

g The diagram contains parallel lines. The marked angles are congruent and alternate exterior angles so the lines are parallel by the converse of the alternate exterior angles theorem.

2

a No, $\angle CBE$ and $\angle FEH$ are corresponding angles, and they're not congruent.

b No, $\angle CBE$ and $\angle BEF$ are consecutive interior angles, and they're not supplementary.

3

a No, $\angle ABC$ and $\angle BCD$ are alternate interior angles, and they're not congruent.

b Yes, $\overline{AB}$ and $\overline{CD}$ form supplementary consecutive interior angles with $\overline{AC}$.

4

a No

b No

c Yes

d No

5 Consecutive interior angles converse

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To prove: $\overleftrightarrow{AC} \parallel \overleftrightarrow{BD}$

	Statements	Reasons
1.	$\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$	Given
2.	$\angle ABD \cong \angle DCA$	Given
3.	$\angle ABD$ and $\angle BDC$ are supplementary	Consecutive interior angles theorem
4.	$\angle DCA$ and $\angle BDC$ are supplementary	Substitution
5.	$\overleftrightarrow{AC} \parallel \overleftrightarrow{BD}$	Consecutive interior angles converse

Additional questions

7 The statement should state that $\angle 1$ and $\angle 2$ are supplementary instead of congruent.

8 $\angle 2$ and 3 are not corresponding angles.

9

a $x = \frac{7}{2}$

b The angle that forms a linear pair with the angle with measure of 136° would have a measure of 44°. If $m\angle 1 = 44^\circ$, $c \parallel d$ by the converse of the alternate interior angles theorem.

10

a $x = 25$

b $x = 3$

c $x = 19$

d $x = 20$

11

a Yes we can conclude $a \parallel b$. It is given that $\angle 1$ and $\angle 14$ are supplementary. $\angle 22$ and $\angle 16$ are supplementary by the linear pair postulate. Since $\angle 22 \cong \angle 1$ and using substitution, $\angle 1$ and $\angle 16$ are supplementary. Therefore, by the congruent supplements theorem, $\angle 14 \cong \angle 16$. By the converse of the corresponding angles postulate, $a \parallel b$.

b No we can not conclude $b \parallel c$. We could conclude $a \parallel b$ by the converse of the consecutive exterior angles theorem since $\angle 10$ and $\angle 7$ are supplementary. However, more information is needed to conclude $b \parallel c$.

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c Yes we can conclude $a \parallel b$. $\angle 19 \cong \angle 2$ by the alternate exterior angles theorem. Using substitution, $\angle 5$ and $\angle 2$ are supplementary. By the corresponding angles postulate $\angle 5 \cong \angle 3$. Using substitution, $\angle 2$ and $\angle 3$ are supplementary. Therefore, by the converse of the consecutive interior angles theorem, $a \parallel b$.

d Yes we can conclude that $a \parallel c$. $\angle 11$ and $\angle 12$ are supplementary using the linear pair postulate. Using the congruent supplements theorem, $\angle 1$ and $\angle 12$ are supplementary. So by the converse of the alternate exterior angles theorem, $a \parallel c$.

e Yes we can conclude $b \parallel c$. $\angle 20 \cong \angle 22$ by the corresponding angles postulate. $\angle 17$ and $\angle 18$ are supplementary by the linear pair postulate. This means that $\angle 17 \cong \angle 22$. By the converse of the alternate interior angles theorem, $b \parallel c$.

f No we can not conclude $a \parallel b$. We can conclude that $b \parallel c$ by the converse of the alternate exterior angles theorem since $\angle 3 \cong \angle 24$.

12

a $\angle BMA, \angle NMS$

b

To prove: $\overleftrightarrow{BS} \parallel \overleftrightarrow{CR}$

	Statements	Reasons
1.	$\angle BMN$ and $\angle QNR$ are supplementary	Given
2.	$\angle BMN$ and $\angle BMA$ are supplementary	Linear pair theorem
3.	$\angle BMA \cong \angle QNR$	Congruent supplement theorem
4.	$\overleftrightarrow{BS} \parallel \overleftrightarrow{CR}$	Alternate exterior angles converse

13

To prove: $\overline{AD} \parallel \overline{BC}$

	Statements	Reasons
1.	$\angle A$ and $\angle D$ are supplementary	Given
2.	$\angle B \cong \angle D$	Given
3.	$\angle A$ and $\angle B$ are supplementary	Substitution
4.	$\overline{AD} \parallel \overline{BC}$	Converse of the consecutive interior angles theorem

14 It is given that $\angle 1$ is supplementary to $\angle 2$. By the linear pair postulate, $\angle 1$ is supplementary

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to $\angle 3$. By the congruent supplements theorem, $\angle 2 \cong \angle 3$. Therefore, by the converse of the corresponding angles postulate, $m \parallel n$.

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To prove: $\angle CST \cong \angle XQP$

	Statements	Reasons
1.	$\overleftrightarrow{AX} \parallel \overleftrightarrow{BY}$	Given
2.	$\overleftrightarrow{BY} \parallel \overleftrightarrow{CZ}$	Given
3.	$\angle XQP \cong \angle YRQ$	Corresponding angles theorem
4.	$\angle YRQ \cong \angle ZSR$	Corresponding angles theorem
5.	$\angle XQP \cong \angle ZSR$	Transitive property of congruence
6.	$\angle ZSR \cong \angle CST$	Vertical angles congruence theorem
7.	$\angle XQP \cong \angle CST$	Transitive property of congruence

16

To prove: $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$

	Statements	Reasons
1.	$\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$	Given
2.	$\angle NBA$ and $\angle ADC$ are supplementary	Given
3.	$\angle NBA$ and $\angle DCR$ are supplementary	Consecutive exterior angles theorem
4.	$\angle ADC \cong \angle DCR$	Congruent supplements theorem
5.	$\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$	Converse of the alternate interior angles theorem

17

To prove: $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$

	Statements	Reasons
1.	$\overleftrightarrow{AB} \parallel \overleftrightarrow{DC}$	Given
2.	$\angle MAB \cong \angle BCQ$	Given
3.	$\angle BCQ \cong \angle ABC$	Alternate interior angles theorem
4.	$\angle MAB \cong \angle ACB$	Transitive property of congruence
5.	$\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$	Converse of the alternate interior angles theorem

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